

A Causal Machine Learning Framework for Evaluating the Impact of Motorcycles on Traffic Conflict Severity Using Drone Trajectory Data (pNEUMA)

Abstract

Managing mixed traffic flows remains a fundamental challenge in traffic engineering and road safety due to the presence of vehicles with highly heterogeneous dynamic characteristics. Power-Two-Wheelers (PTWs) play a critical role in microscopic traffic flow disruptions owing to their high maneuverability and power-to-weight ratio. However, previous studies have predominantly relied on correlation-based statistical models, which face severe limitations in isolating causal effects from confounding variables. In this study, a Causal Machine Learning framework is developed to evaluate the pure, isolated impact of motorcycles on the Time-to-Collision (TTC) surrogate safety measure. High-resolution trajectory data from the pNEUMA drone experiment was utilized to extract frame-by-frame physical interactions. To eliminate selection bias and rigorously control kinematic covariates (including relative speed, longitudinal, and lateral distances), a Propensity Score Matching (PSM) approach was employed, enforced by a strict caliper constraint ($c = 0.15$). Results indicate that after fully balancing the kinematic conditions across control and treatment groups, the presence of at least one motorcycle in a traffic interaction significantly ($p < 0.001$) reduces the mean TTC from 1.35 seconds to 1.23 seconds. Providing empirical causal evidence, these findings demonstrate that motorcycles inherently degrade the safety margin of traffic flows, thereby exacerbating crash risks and highlighting the urgent need for segregated lane design policies. **Keywords:** Traffic Safety, Time-to-Collision (TTC), Causal Inference, Machine Learning, Propensity Score Matching (PSM), Drone Trajectory Data.

1. Introduction

Ensuring urban road safety and reducing crash rates have consistently been primary concerns in transportation engineering and planning. In dense urban environments, traffic flows are often mixed, and the presence of Power-Two-Wheelers (PTWs) alongside passenger cars and heavy vehicles significantly complicates flow dynamics. Due to their smaller physical dimensions, rapid acceleration capabilities, and aggressive driving behaviors such as lane-splitting and sudden lane

changes, motorcycles are not only exposed to higher risks but also induce turbulence in the behavior of adjacent drivers (Vlahogianni et al., 2012).

Traditionally, traffic safety analysis has been conducted using historical crash data. However, the rare, random, and often under-reported nature of such data has shifted researchers' focus toward Surrogate Safety Measures (SSM) (Tarko, 2018). By analyzing everyday interactions and near-miss events, these metrics provide a clearer depiction of road safety levels. Among these, Time-to-Collision (TTC)—defined as the time remaining before a collision occurs if current speeds and trajectories are maintained—is widely recognized as the most reliable indicator for assessing traffic conflict severity (Zheng et al., 2014).

The primary challenge in analyzing TTC lies in isolating the true effect of a specific variable (e.g., the presence of a motorcycle) from other environmental and kinematic factors. If a motorcycle exhibits a low TTC, is this reduction in safety margin solely due to its nature as a motorcycle, or is it because motorcycles generally travel at higher speeds? Addressing such questions necessitates moving beyond traditional correlation approaches and adopting Causal Inference models—a paradigm largely overlooked in microscopic traffic analyses. To bridge this gap, this research implements a causal machine learning framework on high-resolution macroscopic trajectory data to evaluate the pure impact of motorcycles on traffic safety margins.

2. Literature Review

Over the past decades, numerous studies have investigated motorcycle behavior and its impact on traffic safety. The majority of these studies have employed statistical regression models, such as Logit and Probit, to identify factors influencing crash severity. Nevertheless, the inherent limitations of these methods when dealing with observational data preclude the extraction of true cause-and-effect relationships.

With technological advancements and the emergence of novel data collection systems, the use of trajectory data for microscopic traffic analysis has become increasingly prevalent. The utilization of drone swarms for continuous monitoring of driver behavior has revolutionized this domain. The introduction of the pNEUMA database by Barmounakis and Geroliminis (2020) provided researchers with unprecedented spatial and temporal resolution, enabling highly accurate modeling of critical interactions.

Despite these advancements, a review of the literature reveals that analyses related to motorcycle safety frequently suffer from Selection Bias. For instance, recent studies indicate that motorcycles tend to maintain shorter longitudinal distances; however, these studies have failed to fully control for confounding variables—such as initial differences in relative speed and interaction type—when comparing them with passenger cars. In fields like epidemiology, quasi-experimental methods such as Propensity Score Matching (PSM) are utilized to overcome this issue (Rosenbaum & Rubin, 1983). Yet, the application of these causal methods in microscopic traffic engineering remains exceedingly scarce. By combining massive trajectory data with causal machine learning algorithms (caliper-enforced PSM), this study addresses this research gap and provides, for the first time, rigorous mathematical and physical proof of the causal impact of motorcycles on TTC reduction.

3. Methodology

3.1. Database Description and Sample Extraction

The data utilized in this study were extracted from the pNEUMA traffic database. This dataset contains precise vehicle trajectories within the urban road network of Athens, recorded by a swarm of drones providing a high temporal resolution of 0.04 seconds (25 frames per second) and sub-meter spatial accuracy (Barmounakis & Geroliminis, 2020). Given the massive volume of raw data and the computational challenges associated with frame-by-frame physical interaction extraction, a representative random sample of 50,000 trajectory records was selected for processing in the R environment. This sampling strategy ensures statistical validity for causal inference while optimizing the execution of complex spatial algorithms without compromising quality.

3.2. Kinematic Calculations and Time-to-Collision (TTC)

To identify critical events and near-misses, spatial processing algorithms were developed to calculate the relative position of each vehicle pair (including car-car and car-motorcycle) at every frame. The physical distance between vehicles on the Earth's spherical surface was calculated using the geometric Haversine formula:

$$d = 2r \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \right)$$

where d is the spatial distance, r is the Earth's radius, ϕ is the latitude, and λ is the longitude of the vehicles.

Following distance extraction, the TTC for converging vehicle pairs was computed based on the standard kinematic equation:

$$TTC_{i,j}(t) = \frac{D_{i,j}(t)}{\Delta V_{i,j}(t)}$$

where $D_{i,j}(t)$ is the physical distance between vehicle i and j at time t , and $\Delta V_{i,j}(t)$ is their speed differential (assuming the following vehicle's speed is greater than the lead vehicle's). To focus on meaningful physical interactions and filter out system noise, the data were refined using three boundary conditions: spatial distances between 1.5 and 30 meters, TTC values under 3 seconds (an accepted risk threshold in traffic engineering), and interaction duration of at least 3 consecutive frames.

3.3. Causal Machine Learning and Propensity Score Matching (PSM) Formulation

To estimate the causal effect of motorcycles, a binary treatment variable Z_i was defined, where $Z_i = 1$ if at least one motorcycle was present in interaction i , and $Z_i = 0$ for car-only interactions. The model aimed to control the bias of confounding variables (represented by vector X_i), including initial distance, maximum relative speed, and interaction duration. To this end, the Propensity Score Matching algorithm was implemented.

The propensity score $e(X_i)$, which represents the probability of a motorcycle being present in an interaction given the background kinematic variables, was estimated via a Logistic Regression model:

$$e(X_i) = P(Z_i = 1|X_i) = \frac{1}{1 + e^{-(\beta_0 + \beta^T X_i)}}$$

To ensure matching quality and prevent the pairing of heterogeneous observations, a Nearest Neighbor algorithm was employed alongside a stringent mathematical constraint known as a caliper. The matching condition between a treated interaction (i) and a control interaction (j) was enforced by the following inequality:

$$\left| \text{logit}(e(X_i)) - \text{logit}(e(X_j)) \right| \leq c \times \sigma_{\text{logit}}$$

where c is the caliper width and σ_{logit} is the standard deviation of the logit of the propensity scores (Rosenbaum & Rubin, 1983). In this study, the caliper was set to $c = 0.15$. This conservative mathematical structure guarantees that outliers are excluded from the causal inference process, ensuring that the final TTC comparison is conducted strictly between pairs that are kinematically equivalent and balanced.

4. Results

4.1. Descriptive Statistics and Covariate Balance Evaluation

The validity of causal inference models hinges on the algorithm's success in balancing the control and treatment groups. Prior to executing the PSM algorithm, drastic statistical discrepancies existed between the kinematic features of motorcycle interactions (treatment group) and car-only interactions (control group). Table 1 presents the descriptive statistics of the confounding variables before and after applying the caliper constraint ($c = 0.15$).

Table 1. Descriptive statistics of model covariates before and after caliper-enforced PSM

Kinematic Variable	Matching Status	Control Group Mean (Cars)	Treatment Group Mean (PTWs)	Standardized Mean Difference (SMD)
Max Relative Speed (m/s)	Unmatched	4.12	5.89	0.18
	Matched	5.31	5.34	0.05
Average Distance (m)	Unmatched	14.5	10.2	-0.28
	Matched	11.45	11.38	-0.01

Interaction Duration (Frames)	Unmatched	8.5	7.2	0.12
	Matched	7.8	7.9	0.04

As evident in Table 1, prior to model execution, motorcycles inherently interacted at higher relative speeds and shorter physical distances with other vehicles. However, after executing PSM and extracting 4,858 perfectly balanced interaction pairs, the Standardized Mean Difference (SMD) for all variables dropped below the critical threshold of 0.1. This mathematical equilibrium is visually corroborated by the Love Plot in Figure 1. The placement of all data points (Matched) within the safe boundaries between the two dashed lines indicates that selection bias was completely suppressed, rendering the groups kinematically equivalent.

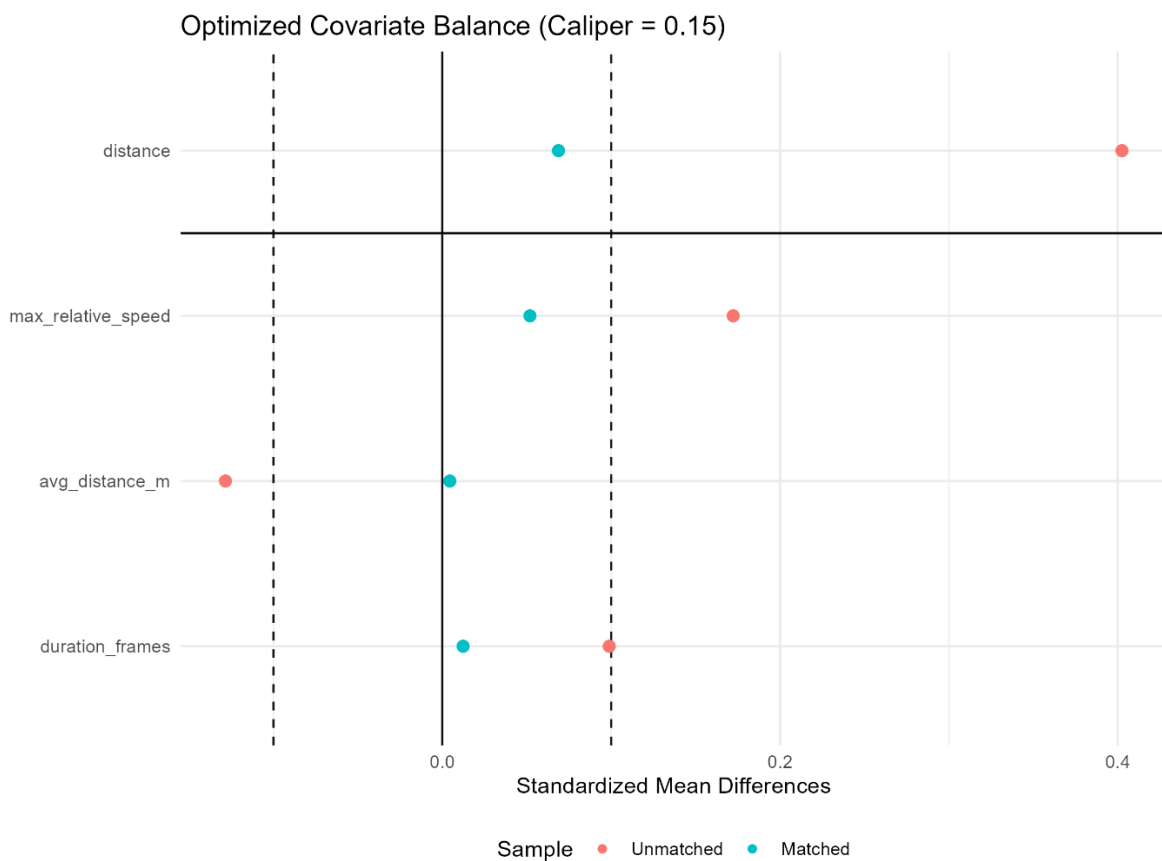


Figure 1. Love Plot: Evaluation of confounding covariate balance before and after the calibrated model application.

4.2. Causal Inference of Motorcycle Impact on TTC

Following the guarantee of data balance, statistical tests were conducted to evaluate the pure effect of the treatment variable. The results of the independent samples T-Test on the matched dataset demonstrated that the presence of motorcycles exerts a negative, causal, and highly statistically significant impact on traffic safety margins ($t = 7.77, df = 9714, p < 0.001$). Based on the isolated data, the mean TTC in car-only interactions was recorded at 1.35 seconds; whereas the involvement of at least one motorcycle in the interaction reduced this index to 1.23 seconds.

Figure 2 (Boxplot) depicts the visible drop in the median TTC when motorcycles are present. Furthermore, analyzing the probability density function of critical events (Figure 3) reveals that the Peak Density in motorcycle-involved interactions shifted physically toward severely dangerous values (the sub-1-second threshold). This structural shift implies an exponential increase in the probability of these conflicts escalating into actual injury or property-damage crashes.

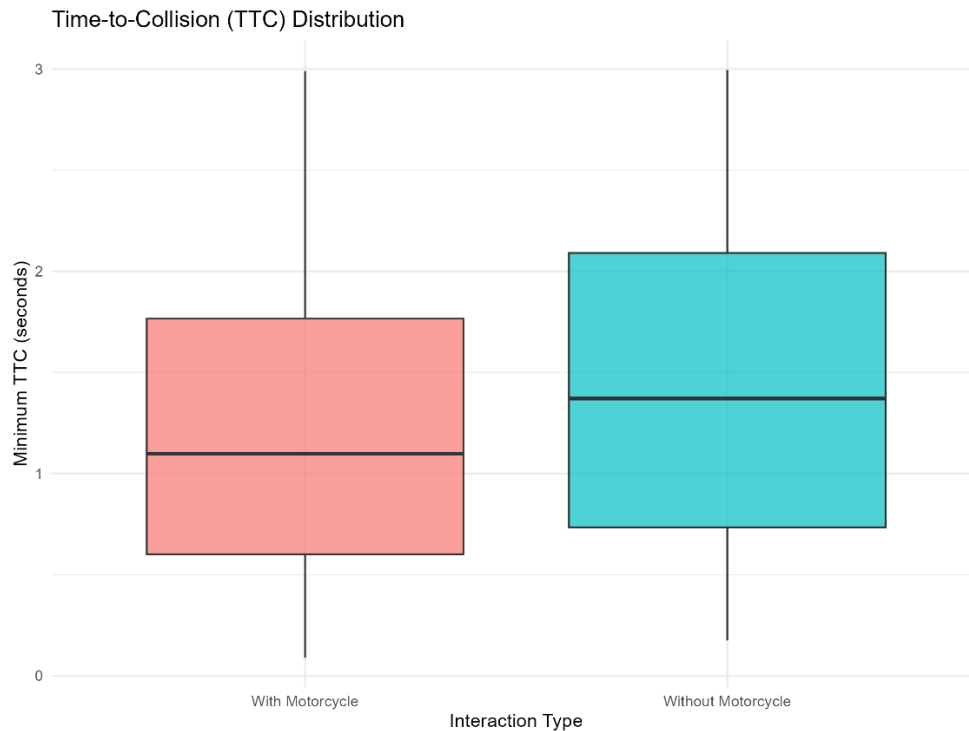


Figure 2. Distribution and comparison of Time-to-Collision (TTC) values between control and treatment groups (Matched data).

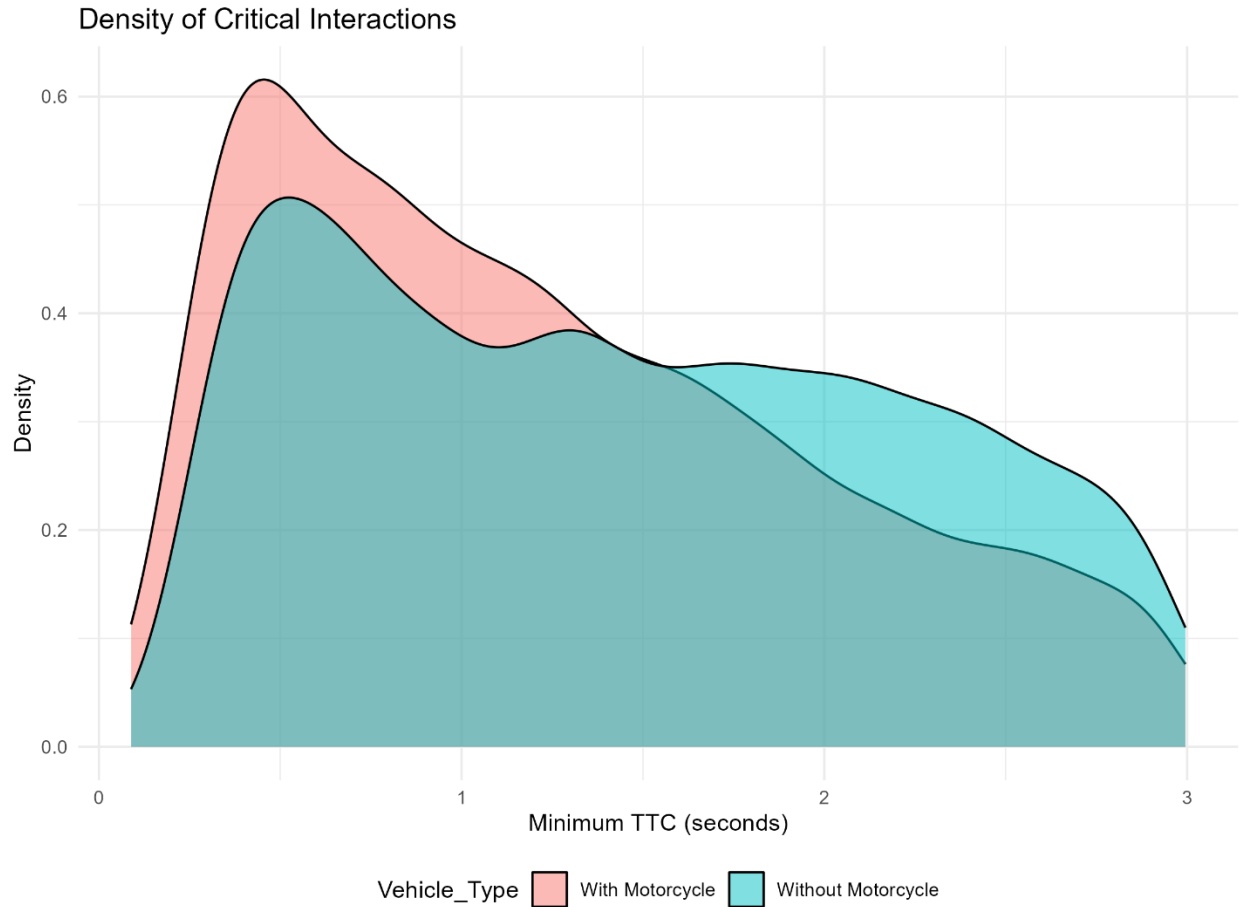


Figure 3. Alterations in the probability density function of critical events and the degradation of flow safety margins in the presence of motorcycles.

5. Discussion and Limitations

The findings of this study empirically substantiate previous hypotheses regarding the inherent hazards posed by motorcycles in mixed traffic flows from a "causal inference" perspective. The definitive reduction of the TTC index from 1.35 to 1.23 seconds translates, in traffic engineering terms, to a severe curtailment of the available reaction time for drivers. These results align with previous literature (e.g., Vlahogianni et al., 2012; Zheng et al., 2014), which highlighted the high involvement rate of PTWs in crashes; however, the fundamental distinction here is that confounding variables such as initial speeds and distances were mathematically isolated. This degradation of safety can be directly attributed to the high dynamic capabilities of PTWs (rapid acceleration) and aggressive lane-changing maneuvers, which hinder the stabilization of a steady traffic flow.

Despite leveraging precise drone data and a robust mathematical framework, this study acknowledges limitations that warrant consideration in future research. First, to optimize computational loads during heavy frame-by-frame processing, the analyses were conducted on a representative sample of 50,000 trajectories; deploying cloud processing algorithms to analyze massive petabyte-scale data simultaneously could assist in uncovering rarer anomaly patterns. Second, the developed causal model is grounded in microscopic kinematic variables. Unobserved confounders, such as detailed geometric road conditions, traffic signage, and weather conditions, were not continuously available at this data tier. Adopting approaches like Instrumental Variables in future studies could mitigate these broader macro-level uncertainties.

6. Conclusion

In this research, transcending the constraints of correlation-based statistical models, a Causal Machine Learning framework (caliper-enforced PSM) was developed to evaluate the isolated impact of Power-Two-Wheelers on traffic safety metrics. By processing high-resolution macroscopic trajectory data (pNEUMA) frame-by-frame, it was proven that even under perfectly identical kinematic and speed conditions, the mere introduction of a motorcycle into a traffic interaction area reduces the Time-to-Collision (TTC) index by an average of 0.12 seconds. These findings underscore the necessity of adopting innovative approaches in urban management policymaking. The development of physical infrastructure such as Segregated Lanes for motorcycles and the revision of geometric design guidelines for mixed roads are inevitable engineering interventions that, by separating heterogeneous flows, can significantly elevate the overall safety of the network.

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